CHAPTER 11

Binomial Identities

Notes By Pr. El Hadiq Zouhair

11.1 Two kinds of proof

Recall $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ counts the k-subsets of an n-set. Binomial identities admit two complementary proof styles: *algebraic* (manipulate factorials, or compare coefficients in the binomial theorem) and *combinatorial* (“double counting”: count one set in two ways). Whenever possible we give both — the combinatorial proof usually explains *why* the identity holds.

11.2 The fundamental identities

Theorem 11.1 (Symmetry)**.** $\displaystyle\binom{n}{k} = \binom{n}{n-k}$.

Proof. Algebraic: both equal $\frac{n!}{k!(n-k)!}$. Combinatorial: selecting k elements is the same as rejecting $n-k$. □

Theorem 11.2 (Pascal)**.** $\displaystyle\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$.

Proof. Fix a distinguished element x of an $(n+1)$-set: k-subsets containing x number $\binom{n}{k-1}$, those avoiding x number $\binom{n}{k}$. □

Pascal’s triangle hides many patterns: the hockey-stick sums below, the Fibonacci numbers along shallow diagonals ($\sum_k \binom{n-k}{k} = F_{n+1}$), the parity fractal of Sierpiński, and the central column $\binom{2n}{n}$ governing Catalan numbers.

Theorem 11.3 (Absorption)**.** $\displaystyle k\binom{n}{k} = n\binom{n-1}{k-1}$, equivalently $\binom{n}{k} = \frac{n}{k}\binom{n-1}{k-1}$.

Proof. Count pairs (committee of k, its chair) from n people in two ways: choose the committee then the chair ($\binom{n}{k} \cdot k$), or the chair then the rest ($n \cdot \binom{n-1}{k-1}$). □

11.3 The binomial theorem and its harvest

Theorem 11.4 (Binomial theorem)**.** $(x+y)^n = \displaystyle\sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$

Substituting values of x, y yields identities mechanically.

Theorem 11.5 (Row sum)**.** $\displaystyle\sum_{k=0}^{n} \binom{n}{k} = 2^n.$

Proof. Three proofs. (1) Set $x = y = 1$ in the binomial theorem. (2) Combinatorial: both sides count subsets of an n-set, the left side by size. (3) Induction via Pascal's identity. □

Theorem 11.6 (Alternating sum)**.** $\displaystyle\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0$ for $n \geq 1$ — set $x = 1$, $y = -1$. Hence even-sized subsets equal odd-sized subsets in number (2^{n-1} each).

Theorem 11.7 (Weighted sums)**.**

$$\sum_{k=0}^{n} 2^k \binom{n}{k} = 3^n, \qquad \sum_{k=0}^{n} k \binom{n}{k} = n\, 2^{n-1}, \qquad \sum_{k=0}^{n} k^2 \binom{n}{k} = n(n+1)\, 2^{n-2}.$$

Proof. First: $x = 1, y = 2$. Second: absorption gives $\sum_k k\binom{n}{k} = n \sum_k \binom{n-1}{k-1} = n\, 2^{n-1}$; combinatorially, count (subset, distinguished member) pairs. Third: write $k^2 = k(k-1) + k$ and apply absorption twice: $\sum k(k-1)\binom{n}{k} = n(n-1)2^{n-2}$, then add $n\, 2^{n-1}$ to get $n(n-1)2^{n-2} + n\, 2^{n-1} = n(n+1)2^{n-2}$. □

11.4 Vandermonde and friends

Theorem 11.8 (Vandermonde)**.** $\displaystyle\binom{m+n}{r} = \sum_{k=0}^{r} \binom{m}{k}\binom{n}{r-k}.$

Example 11.9. A jury of 4 from 6 magistrates and 5 citizens: $\binom{11}{4} = 330$ directly; by cases on magistrates: $\binom{6}{0}\binom{5}{4} + \binom{6}{1}\binom{5}{3} + \binom{6}{2}\binom{5}{2} + \binom{6}{3}\binom{5}{1} + \binom{6}{4}\binom{5}{0} = 5 + 60 + 150 + 100 + 15 = 330$. ✓

Theorem 11.10 (Sum of squares)**.** $\displaystyle\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}$ — Vandermonde with $m = n = r$, using symmetry $\binom{n}{n-k} = \binom{n}{k}$.

Theorem 11.11 (Hockey stick). $\displaystyle\sum_{i=r}^{n}\binom{i}{r}=\binom{n+1}{r+1}$.

Proof. Classify $(r+1)$-subsets of $\{1,\dots,n+1\}$ by largest element $i+1$: the remaining r elements come from $\{1,\dots,i\}$, in $\binom{i}{r}$ ways, for $i=r,\dots,n$. $\square$

11.5 Extensions

Definition 11.12 (Generalised binomial coefficient). For real α and $k\in\mathbb{N}$: $\binom{\alpha}{k}=\frac{\alpha(\alpha-1)\cdots(\alpha-k+1)}{k!}$. Newton's binomial series $(1+x)^\alpha=\sum_k\binom{\alpha}{k}x^k$ converges for $|x|<1$; the *negative-binomial* case gives $\binom{-n}{k}=(-1)^k\binom{n+k-1}{k}$, linking to stars and bars.

Definition 11.13 (Catalan numbers). $C_n=\dfrac{1}{n+1}\dbinom{2n}{n}=\dbinom{2n}{n}-\dbinom{2n}{n+1}$: $1,1,2,5,14,42,\dots$ They count balanced parenthesisations, monotonic lattice paths not crossing the diagonal, triangulations of convex polygons, and binary trees.

The multinomial coefficients $\binom{n}{n_1,\dots,n_m}$ extend the whole theory from two blocks to m; and *generating functions* — viewing $\sum_k\binom{n}{k}x^k=(1+x)^n$ as a polynomial identity — turn combinatorial convolutions into products of series, the method behind the Vandermonde proof and far beyond.

11.6 The combinatorial-proof template

To prove LHS = RHS combinatorially: (1) pose a concrete counting question; (2) show the LHS answers it one way; (3) show the RHS answers the same question another way; (4) conclude the counts are equal. Practise the template on the worked example below.

Example 11.14 (Portfolio choice). Maeve must pick a team of k analysts from n candidates and name one team lead, but she may also count by first naming the lead (n ways) and then completing the team ($\binom{n-1}{k-1}$ ways). The two counts, $k\binom{n}{k}$ and $n\binom{n-1}{k-1}$, must agree: absorption, proved by the template.

11.7 Master cheat sheet

Symmetry	$\binom{n}{k} = \binom{n}{n-k}$
Pascal	$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$
Absorption	$k\binom{n}{k} = n\binom{n-1}{k-1}$
Row sum	$\sum_k \binom{n}{k} = 2^n$
Alternating	$\sum_k (-1)^k \binom{n}{k} = 0 \quad (n \geq 1)$
Weighted	$\sum_k k\binom{n}{k} = n2^{n-1}, \quad \sum_k k^2\binom{n}{k} = n(n+1)2^{n-2}$
Powers of 3	$\sum_k 2^k \binom{n}{k} = 3^n$
Vandermonde	$\binom{m+n}{r} = \sum_k \binom{m}{k}\binom{n}{r-k}$
Squares	$\sum_k \binom{n}{k}^2 = \binom{2n}{n}$
Hockey stick	$\sum_{i=r}^{n} \binom{i}{r} = \binom{n+1}{r+1}$
Catalan	$C_n = \frac{1}{n+1}\binom{2n}{n}$

Common pitfalls: forgetting that $\binom{n}{k} = 0$ for $k < 0$ or $k > n$; shifting summation indices incorrectly; applying symmetry inside a sum without adjusting the range; and assuming an identity for real upper index when it only holds for integers.

11.8 Summary

Every identity in this chapter flows from three sources: Pascal's recurrence, the binomial theorem with clever substitutions, and double counting. Master the cheat sheet, but more importantly master the *template*: a binomial identity is an invitation to find the single question that both sides answer.

Exercises

Exercise 11.1 EASY

Evaluate without a calculator: $\binom{10}{8}$, $\binom{12}{1}$, $\binom{9}{4} + \binom{9}{5}$.

Exercise 11.2 EASY

Use the row-sum and alternating-sum identities to find $\sum_{k \text{ even}} \binom{n}{k}$ for $n \geq 1$.

Exercise 11.3 EASY

Verify the hockey-stick identity $\binom{2}{2} + \binom{3}{2} + \binom{4}{2} + \binom{5}{2} = \binom{6}{3}$ numerically.

Exercise 11.4 MEDIUM

Give a combinatorial proof of $\binom{n}{2}\binom{n-2}{k-2} = \binom{n}{k}\binom{k}{2}$ $(2 \leq k \leq n)$.

Exercise 11.5 MEDIUM

Evaluate $\sum_{k=0}^{n} 3^k \binom{n}{k}$ and $\sum_{k=0}^{n}(-2)^k \binom{n}{k}$.

Exercise 11.6 MEDIUM

Prove $\sum_{k=1}^{n} k\binom{n}{k} = n2^{n-1}$ combinatorially, by counting committees with a chair.

Exercise 11.7 MEDIUM

Using Vandermonde, evaluate $\sum_{k=0}^{m} \binom{m}{k}\binom{n}{p+k}$ as a single binomial coefficient ($p + m \leq n$).

Exercise 11.8 MEDIUM

Show that $\binom{2n}{n}$ is even for $n \geq 1$. *Hint:* $\binom{2n}{n} = \binom{2n-1}{n-1} + \binom{2n-1}{n}$ and symmetry.

Exercise 11.9 HARD

Prove the Fibonacci diagonal identity $\sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k} = F_{n+1}$ (with $F_1 = F_2 = 1$), by induction using Pascal's identity.

Exercise 11.10 HARD

Prove $\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}$ combinatorially, by counting n-subsets of a set of n women and n men.

Exercise 11.11 HARD

Show $C_n = \binom{2n}{n} - \binom{2n}{n+1}$ counts lattice paths from $(0, 0)$ to (n, n) (unit right/up steps) that never rise above the diagonal, via the reflection principle.

Solutions to the Exercises

Solution 11.1. $\binom{10}{8} = \binom{10}{2} = 45$; $\binom{12}{1} = 12$; $\binom{9}{4} + \binom{9}{5} = \binom{10}{5} = 252$ (Pascal). ∎

Solution 11.2. Adding $\sum_k \binom{n}{k} = 2^n$ and $\sum_k(-1)^k\binom{n}{k} = 0$ cancels odd k and doubles even k: $\sum_{k \text{ even}} \binom{n}{k} = 2^{n-1}$. ∎

Solution 11.3. $1 + 3 + 6 + 10 = 20 = \binom{6}{3}$. ✓ ∎

Solution 11.4. Both sides count (committee of k from n people, with 2 designated co-chairs): left side picks the co-chairs first ($\binom{n}{2}$), then fills the committee ($\binom{n-2}{k-2}$); right side picks the committee ($\binom{n}{k}$), then its co-chairs ($\binom{k}{2}$). ∎

Solution 11.5. Binomial theorem with $(x, y) = (1, 3)$: 4^n. With $(1, -2)$: $(-1)^n$. ∎

Solution 11.6. Count pairs (committee S of any size, chair $c \in S$) from n people. By size: choose size k, the committee, then the chair: $\sum_k \binom{n}{k}k$. Chair first: n choices, then any subset of the remaining $n - 1$ people joins: $n\,2^{n-1}$. Equate. ∎

Solution 11.7. By symmetry $\binom{m}{k} = \binom{m}{m-k}$, the sum is $\sum_k \binom{m}{m-k}\binom{n}{p+k}$, a Vandermonde convolution picking $m + p$ objects from $m + n$: $\binom{m+n}{m+p}$. ∎

Solution 11.8. Pascal: $\binom{2n}{n} = \binom{2n-1}{n-1} + \binom{2n-1}{n}$, and the two terms are equal by symmetry ($(n-1)+n = 2n-1$). Hence $\binom{2n}{n} = 2\binom{2n-1}{n-1}$ is even. ■

Solution 11.9. Let $S_n = \sum_k \binom{n-k}{k}$. Check $S_1 = 1 = F_2$, $S_2 = 2 = F_3$. For the step, Pascal gives $\binom{n-k}{k} = \binom{n-1-k}{k} + \binom{n-1-k}{k-1}$; summing over k, the first parts sum to S_{n-1} and the second parts re-index ($k \to k-1$) to S_{n-2}. So $S_n = S_{n-1} + S_{n-2}$ — the Fibonacci recurrence with Fibonacci initial values, hence $S_n = F_{n+1}$ for all n. ■

Solution 11.10. Choosing n people from n women + n men splits by the number k of women chosen: $\binom{n}{k}$ ways for the women and $\binom{n}{n-k} = \binom{n}{k}$ for the men, giving $\sum_k \binom{n}{k}^2$. Direct count: $\binom{2n}{n}$. Equate. ■

Solution 11.11. All paths to (n,n): $\binom{2n}{n}$. A "bad" path touches the line $y = x+1$; reflecting its prefix up to the first touch across that line bijects bad paths with paths from $(-1,1)$ to (n,n), which number $\binom{2n}{n+1}$. Good paths: $\binom{2n}{n} - \binom{2n}{n+1} = C_n$. ■